# Definition-Only RAG for Financial QA: Average Gains and a Diagnostic Boundary

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## Introduction

**The failure mode.** A model can miss a specialized question for no reason other than not knowing what one term means. Qwen-2.5-7B answers *American Depositary Receipt* where the answer is *Guilder Share*. One sentence of Investopedia flips it.

**Scale has not removed it.** On MMLU finance subjects a frontier model still misses **one question in eight**; 7–8B models miss **40–50%**.

**RQ1** **Does injecting a retrieved definition improve financial QA on average?**
**RQ2** **Can the model's own term knowledge say, per question, whether it is needed?**

**Why a glossary definition.** It is the smallest and most auditable payload a retriever can inject, and the only one whose content can be checked term by term.

**Two operational levels.** One signal, term knowledge: a diagnostic of what a model holds, and the control for a selective retrieval policy.

## Related work

**Definition modeling**

probes what a model has internalized about a domain.

- Noraset et al. 2017; Gardner et al. 2022
- Finance: Jhirad et al. 2023. Multilingual: Marrese-Taylor et al. 2025

**Selective and adaptive RAG**

assumes some observable property of the question or model state carries enough signal to route between retrieve and no-retrieve.

- Lewis et al. 2020; Asai et al. 2024; Jeong et al. 2024

**Uncertainty-style routing baselines**

Semantic entropy (Farquhar et al. 2024) and option-margin signals.

- Wang et al. 2024: first-token MCQ scoring can disagree with the model's final text answer.

We test the **narrowest version** of that assumption in finance, using definition-level term knowledge, the most semantically transparent candidate signal this domain offers.

## Method and Data

**Diagnostic.** The model writes the definition itself. Measures what it already holds, no retrieval.

**Interventional.** We supply gold definitions through retrieval and measure what changes downstream.

**Model panel.** Qwen-2.5-7B and 14B, Llama-3.1-8B, Mistral-Small-24B, GPT-5.2: four families from 7B to a frontier closed model, so a capability-dependent effect is separable from a universal one.

**Retrieval pipeline.** A lexical retriever and a dense sentence-encoder each propose candidates from a fixed glossary of **13,609** Investopedia terms, the two ranked lists are fused, a cross-encoder reranks them, and the **top-3** snippets are injected above the question stem. Option letters are scored by log-probability at temperature 0, fixed across every experiment.

**Paired evaluation.** Every question is answered twice, with the definitions and without, so each lands in one of four states: correct in both, **fixed**, **broken**, or wrong in both. Net gain is the fix rate minus the break rate, and an oracle that retrieved only where it helps would collect the fixes without paying the breaks.

| Resource | Size | Role |
|---|---|---|
| Investopedia glossary | 13,609 | index + reference |
| MMLU Finance | 1,124 q | primary benchmark |
| XFinBench TU | 1,790 q | replication |
| Term annotation | 995 q | per-question map |
| Invented terms | 50 | memorisation control |

**Table 1.** The glossary, the two benchmarks, and the resources derived from them.

**Per-question term annotation.** No MMLU–Investopedia mapping existed, so we built one. Two LLM annotators proposed candidates (**κ = 0.76**); the first author adjudicated all **2,233** pairs by hand, leaving **995** term-bearing questions.

## Results

**Definition-only retrieval helps on average,** and the gain is confined to a mid-capability band on **both** benchmarks.

| Model | Base | +RAG | Δ |
|---|---|---|---|
| *MMLU Finance (n = 1,124)* | | | |
| Qwen-2.5-7B | 60.8 | 64.1 | **+3.3** |
| Qwen-2.5-14B | 68.2 | 68.1 | −0.1 |
| Llama-3.1-8B | 52.1 | 57.6 | **+5.4** |
| Mistral-S-24B | 72.3 | 72.5 | +0.2 |
| GPT-5.2 | 87.8 | 88.1 | +0.3 |
| *XFinBench TU (n = 1,790)* | | | |
| Qwen-2.5-7B | 70.5 | 75.3 | **+4.8** |
| Qwen-2.5-14B | 77.7 | 76.8 | −0.8 |
| Llama-3.1-8B | 61.5 | 63.5 | **+2.0** |
| Mistral-S-24B | 78.9 | 78.9 | ±0.0 |
| GPT-5.2 | 87.6 | 88.5 | +0.9 |

**Table 2.** Accuracy with and without definition-only RAG. Coloured Δ: 95% CI excludes zero (paired bootstrap, 2,000 resamples).

**The split follows neither size nor family.** No stalling model gains even in its own weakest term-knowledge quartile.

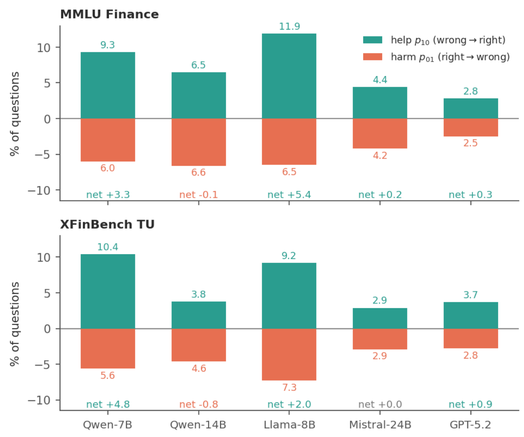



**Figure 1.** Per-model paired-removal decomposition on both benchmarks.

**The gain is the definition, not leakage.** Only 20 of 3,228 injected snippets contain a full answer string, and removing them changes nothing.

| Condition | Acc. | vs RAG |
|---|---|---|
| No RAG (baseline) | 60.8 | −2.9 |
| RAG (as reported) | 63.7 | — |
| drop answer-matching | 63.7 | ±0.0 |
| mask all term names | 61.9 | −1.8 |
| *Invented terms (n = 50, unmemorisable)* | | |
| GPT-5.2 | 36.0 → | 100.0 |
| Qwen-2.5-7B | 30.0 → | 100.0 |

**Table 3.** Controls on the source of the gain.

**A positive control.** On unmemorisable terms the frontier model uses a definition as completely as the mid-capability one, so its flat Δ is not an inability to read one.

## Analysis

**A net of zero hides two-sided movement.** Qwen-2.5-14B sits at Δ ≈ 0 while retrieval still changes 147 of 1,124 answers.

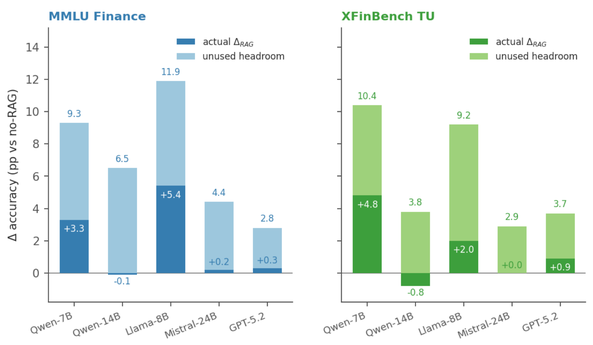



**Figure 2.** Actual Δ against the implementable-oracle upper bound.

**Term knowledge does not predict per question.** All 40 cells lie inside $|r| < 0.06$ ($p = 0.91$).

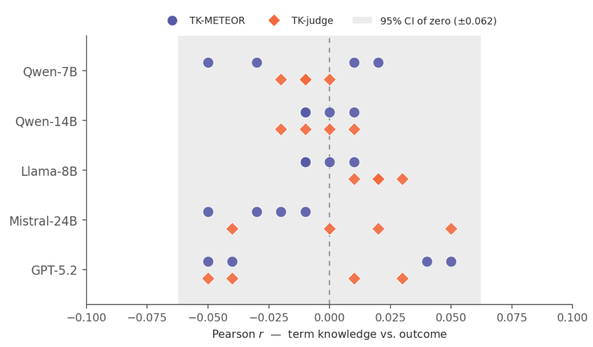



**Figure 3.** Per-question correlations as a forest plot.

**No cheap internal signal approaches the oracle.**

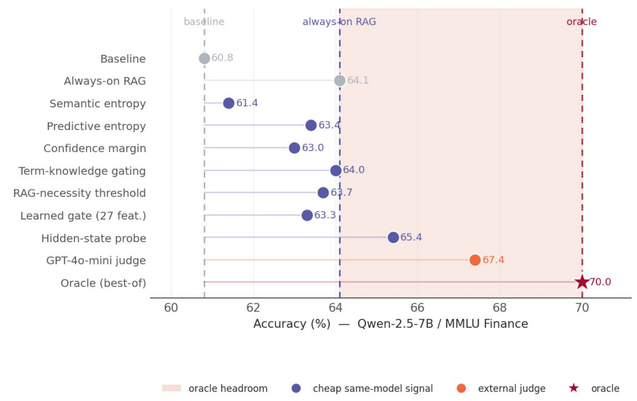



**Figure 4.** Routing-signal accuracies against the oracle gap.

**What the three panels say.** Eight signals, each swept in its own favour; only the external judge beats always-on RAG (67.4 vs 64.1, oracle 70.0), and it is itself a second and stronger model. Filtering which definitions to inject fails too. The injected text perturbs the scores assigned to the four options, and whether that helps or hurts depends on question, option set and snippet together.

## References

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**Diagnosis and intervention come apart: an aggregate correlation over models licenses neither per-instance use of the same probe nor the routing built on it.**

github.com/youngandlei/definition-only-rag